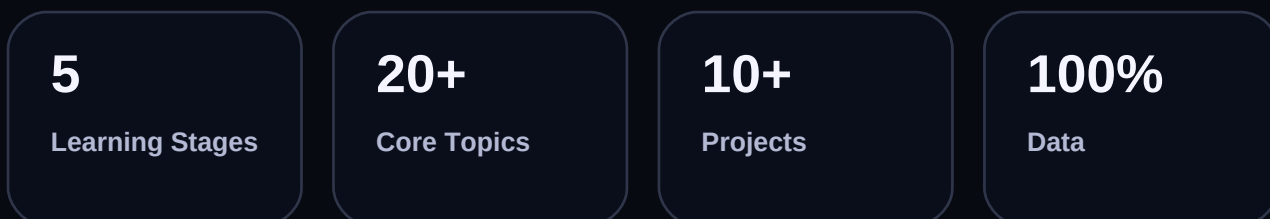
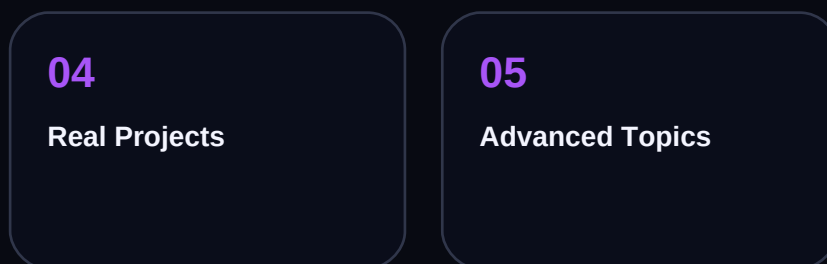
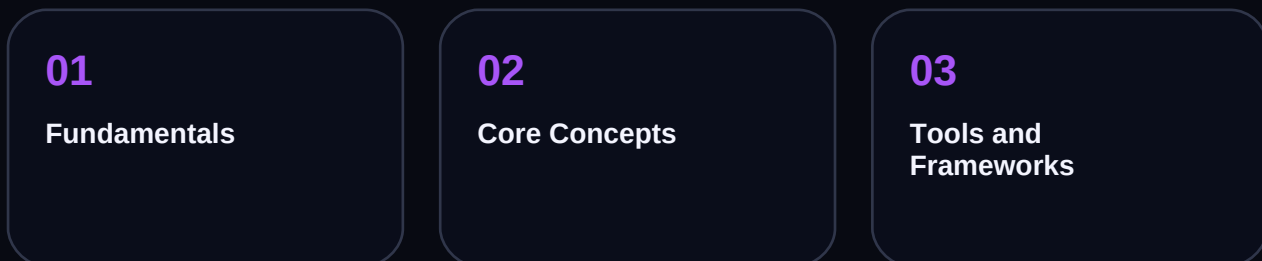
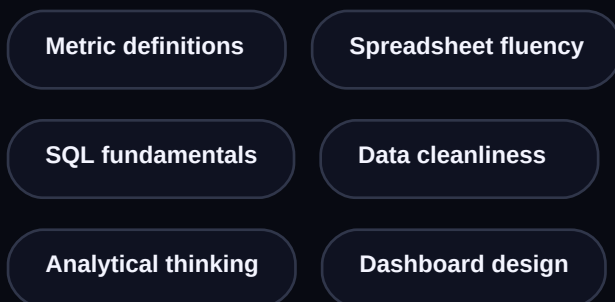




DATA ANALYTICS

Complete Learning Roadmap

Deliver practical business intelligence with trusted metrics, dashboards, and reporting.



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Metric definitions • Spreadsheet fluency • SQL fundamentals • Data cleanliness

02

Core Concepts

Analytical thinking • Dashboard design • Funnel analysis • Cohort analysis

03

Tools and Frameworks

Power BI/Tableau

04

Real Projects

Executive dashboard • Retention analysis • Revenue report automation • Data QA checklist

05

Advanced Topics

Semantic layer design • Self-serve analytics governance • Advanced experimentation analytics • Forecasting basics

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.



Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Metric definitions
- Spreadsheet fluency
- SQL fundamentals
- Data cleanliness

Tools & Frameworks

- Metric definitions
- Spreadsheet fluency
- SQL fundamentals
- Data cleanliness

Recommended Projects

- Data Analytics workshop using Metric definitions
- Data Analytics portfolio project with Spreadsheet fluency

Stage 1 Milestone Complete Data Analytics workshop using Metric definitions and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Analytical thinking
- Dashboard design
- Funnel analysis
- Cohort analysis

Tools & Frameworks

- Analytical thinking
- Dashboard design
- Funnel analysis
- Cohort analysis

Recommended Projects

- Data Analytics workshop using Analytical thinking
- Data Analytics portfolio project with Dashboard design

Stage 2 Milestone Complete Data Analytics workshop using Analytical thinking and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Power BI/Tableau
- dbt basics
- Warehouse querying
- Data visualization tooling

Tools & Frameworks

- Power BI/Tableau

Recommended Projects

- Data Analytics workshop using Power BI/Tableau
- Data Analytics portfolio project with dbt basics

Stage 3 Milestone Complete Data Analytics workshop using Power BI/Tableau and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Executive dashboard
- Retention analysis
- Revenue report automation
- Data QA checklist

Tools & Frameworks

- Executive dashboard
- Retention analysis
- Revenue report automation
- Data QA checklist

Recommended Projects

- Data Analytics workshop using Executive dashboard
- Data Analytics portfolio project with Retention analysis

Stage 4 Milestone Complete Data Analytics workshop using Executive dashboard and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

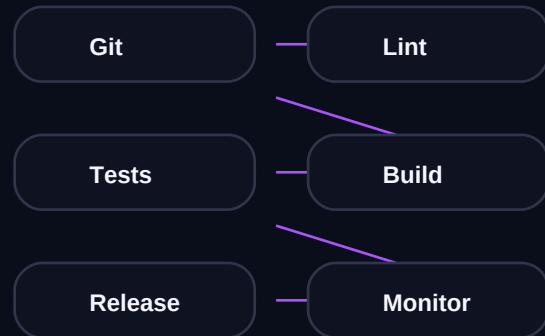
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Semantic layer design
- Self-serve analytics governance
- Advanced experimentation analytics
- Forecasting basics

CI/CD Pipeline



Stage 5 Milestone

A production-grade data analytics workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Data Analytics docs and community guides

Link: <https://learn.microsoft.com/power-bi/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/data-analyst>

PAID

Structured specialization track focused on Data Analytics

Link: <https://roadmap.sh/data-analyst>

FREE

Data Analytics roadmap reference

Link: <https://roadmap.sh/data-analyst>

FREE

Official documentation

Link: <https://learn.microsoft.com/power-bi/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Data Analytics?
- Which project best proves Stage 4 competency in Data Analytics?
- What trade-offs appear in Stage 5 decisions for Data Analytics?